Last Time:

- V, W v.s. A fcn map $T: V \to W$ is a linear transformation (or lin map / homomorphism) if
  (1) $T(v_1 + v_2) = T(v_1) + T(v_2) \quad \forall v_1, v_2 \in V$
  (2) $T(cv) = cT(v) \quad \forall v \in V \ \& \ \forall c \in \mathbb{R}$.
- $\mathcal{L}(V,W) :=$ set of all lin maps $V \to W$
  $\mathcal{L}(V) := \mathcal{L}(V, V)$
- $T \in \mathcal{L}(V,W)$ then $T(0) = 0$ and $T(-v) = -T(v)$ $\forall v \in V$.

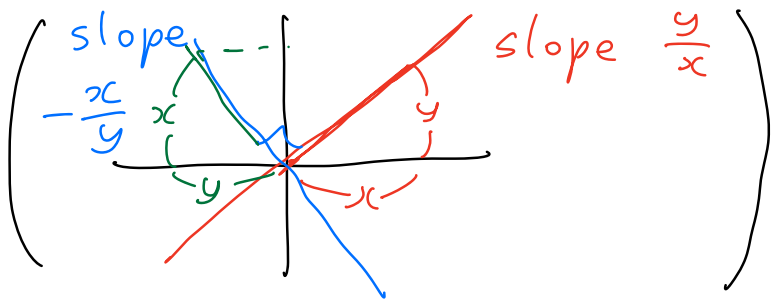
Today:

Continue §2.1 Linear Maps: ker, im, isom (Exs)

Ex 2.3:

(1) Rotation by $\frac{\pi}{2}$ CCW in $\mathbb{R}^2$ is def by

$$R: \mathbb{R}^2 \to \mathbb{R}^2, \quad \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} -y \\ x \end{bmatrix}$$



This is a lin map.

Pf: WTS: Check conds of lin map (Def 2.1)

$\forall \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \in \mathbb{R}^2$ and $\forall c \in \mathbb{R}$,

(1): $R\left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}\right) = R\left(\begin{bmatrix} x_1+x_2 \\ y_1+y_2 \end{bmatrix}\right) = \begin{bmatrix} -(y_1+y_2) \\ x_1+x_2 \end{bmatrix}$

$= \begin{bmatrix} -y_1 \\ x_1 \end{bmatrix} + \begin{bmatrix} -y_2 \\ x_2 \end{bmatrix} = R\left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}\right) + R\left(\begin{bmatrix} x_2 \\ y_2 \end{bmatrix}\right)$ ☒

(2): $R\left(c\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}\right) = R\left(\begin{bmatrix} cx_1 \\ cy_1 \end{bmatrix}\right) = \begin{bmatrix} -cy_1 \\ cx_1 \end{bmatrix} = c\begin{bmatrix} -y_1 \\ x_1 \end{bmatrix}$

$= cR\left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}\right)$. □

(2) Fix $a, b, c \in \mathbb{R}$ and consider a map

$T: \mathbb{R}^3 \to \mathbb{R}$. Then T is a lin map

$(x, y, z) \mapsto ax + by + cz$

Pf: Exercise (Similar to Ex 2.3(1))

(3) (Calc I) $D: \mathbb{R}[x]_{\le n} \longrightarrow \mathbb{R}[x]_{\le n-1}$ differentiation

$f \longmapsto f'$ is a lin map,

b/c $(f+g)' = f' + g'$ and $(cf)' = cf'$ for $c \in \mathbb{R}$

and $f, g \in \mathbb{R}[x]_{\le n}$.

Take this granted: (Calc I: $D\left(\sum_{m=0}^{n} c_m x^m\right) = \sum_{m=1}^{n} m c_m x^{m-1} \overset{u=m-1}{=} \sum_{u=0}^{n-1} (u+1) c_{u+1} x^u$)

(4) Identity map (id) on a v.s. V is $I_V = \mathbb{1}_V : V \longrightarrow V$, $v \longmapsto v$,

which is a lin operator (op) on V (i.e., $I_V \in \mathcal{L}(V) = \mathcal{L}(V, V)$)

Pf: $\forall v, w \in V$ and $\forall c \in \mathbb{R}$,

(1): $I_V(v+w) = v+w = I_V(v) + I_V(w)$

(2): $I_V(cv) = cv = c\, I_V(v)$

$\Rightarrow I_V \in \mathcal{L}(V)$ by Def 2.1 $\square$

(5) Zero map on v.s. V, W is $T_0 : V \longrightarrow W$, $v \longmapsto 0$ which

is a lin map. Abuse of notation: $0 \in \mathcal{L}(V,W)$ means T_0. (Pf: Exercise: Similar to Ex 2.3(4))

Exercise: $T \in \mathcal{L}(V,W)$, then

$T\left(\sum_{i=1}^{n} a_i v_i\right) = \sum_{i=1}^{n} a_i T(v_i)$ $\forall a_1, \dots, a_n \in \mathbb{R}$ and

$\forall v_1, \dots, v_n \in V$. (Hint: use Induction on n)

Rmk 2.4: Similar to how $\mathcal{F}(X, \mathbb{R})$ is a v.s.,

$\mathcal{L}(V,W)$ equipped w/ $+, \cdot$ def by

$+ : \mathcal{L}(V,W) \times \mathcal{L}(V,W) \longrightarrow \mathcal{L}(V,W)$ and

$T, U \longmapsto \left(T+U : V \longrightarrow W,\ v \longmapsto T(v)+U(v)\right)$

$\cdot : \mathbb{R} \times \mathcal{L}(V,W) \longrightarrow \mathcal{L}(V,W)$

$c, T \longmapsto \left(cT : V \longrightarrow W,\ v \longmapsto cT(v)\right)$

is a v.s.

Pf: Exercise (Similar to Ex 1.8 for $\mathcal{F}(X,\mathbb{R})$). □
(also need to prove why $T+U$ & cT are lin maps)

Kernel and Image

Motivation: Given a fcn $f:\mathbb{R}\to\mathbb{R}$, we are interested in finding roots ($f^{-1}(0)$) and image ($f(\mathbb{R})$) based on High School Math. Let's do this for lin maps!

Def 2.5: $T\in\mathcal{L}(V,W)$. Then def

(1) the kernel (ker) (or null space (sp)) (FIS) of T as

$$\ker(T) := \{v\in V : T(v)=0\} \ (=T^{-1}(0) \text{ preimg of } 0) \subseteq V$$

(2) the image (im(g)) (or range (FIS)) of T as

$$\operatorname{im}(T) := \{w\in W : \exists\, v\in V \text{ s.t. } w=T(v)\} \ (=T(V)) \subseteq W$$

Prop 2.6: $T\in\mathcal{L}(V,W)$. Then $\ker(T)$ is a subsp of V, and $\operatorname{im}(T)$ is a subsp of W.

Pf: For $\ker(T)\subseteq V$, need to check 3 conds of Subsp Crit.

(1): $T(0)\overset{\text{Rmk 2.2}}{=}0 \Rightarrow 0\in T^{-1}(0)=\ker(T)$.

(2): $\forall\, v_1,v_2\in\ker(T)$, $T(v_1+v_2)\overset{T \text{ lin}}{=}T(v_1)+T(v_2)=0+0=0$

$\Rightarrow v_1+v_2\in\ker(T)$.

(3): $\forall v \in \ker(T)$ and $\forall c \in \mathbb{R}$,

$$T(cv) \overset{T \text{ lin}}{=} cT(v) = c \cdot 0 = 0 \Rightarrow cv \in \ker(T).$$

Thus, Subsp Crit $\Rightarrow \ker(T)$ is a subsp of V. ▨

For $\operatorname{im}(T) \subseteq W$, check 3 conds of Subsp Crit:

(1): $0 \overset{\text{Rmk } 2.2}{=} T(0) \in T(V) = \operatorname{im}(T)$

(2): $\forall w_1, w_2 \in \operatorname{im}(T)$, $\exists v_1, v_2 \in V$ s.t. $w_i = T(v_i)$ $\forall i$ by def of $\operatorname{im}(T)$. Then

$$w_1 + w_2 = T(v_1) + T(v_2) \overset{T \text{ lin}}{=} T(v_1 + v_2) \in \operatorname{im}(T).$$

(3): $\forall w \in \operatorname{im}(T)$, $\exists v \in V$ s.t. $w = T(v)$. Then $\forall c \in \mathbb{R}$,

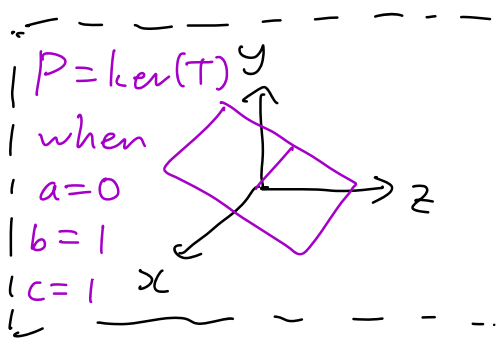
$$cw = cT(v) \overset{T \text{ lin}}{=} T(cv) \in \operatorname{im}(T).$$

Thus, Subsp Crit $\Rightarrow \operatorname{im}(T)$ is a subsp of W. □

Ex 2.7: Recall from Ex 2.3(3) that given $a, b, c \in \mathbb{R}$, then $T : \mathbb{R}^3 \to \mathbb{R}$, $(x, y, z) \mapsto ax + by + cz$ is a lin map.

Then $\ker(T) = \{(x, y, z) \in \mathbb{R}^3 : ax + by + cz = 0\}$ is a subsp of $\mathbb{R}^3$ by Prop 2.6.

(Later: $\ker(T)$ is a plane (dim 2) iff $(a, b, c) \neq 0 \in \mathbb{R}^3$)



inj
injective

Prop 2.8: $T \in \mathcal{L}(V,W)$. T is 1-1 $\iff \ker(T) = \{0\}$.

Pf: ($\Rightarrow$): If T inj and $T(v) = 0$, then $v = 0$ b/c $T(0) = 0$ (as T lin). Thus, $\ker(T) = \{0\}$. ▨

($\Leftarrow$): Assume $\exists\, x, y \in V$ s.t. $T(x) = T(y)$. Then

$T(x-y) = T(x) - T(y) = T(x) - T(x) = 0$ by assump

$\Rightarrow x - y \in \ker(T) = \{0\} \Rightarrow x - y = 0 \Rightarrow x = y$.

Thus, T is inj b/c $\nexists\, x \neq y$ in V w/ $T(x) = T(y)$.

□

Prop 2.9: $T \in \mathcal{L}(V,W)$, $S = \{v_1, \dots, v_n\} \subseteq V = \text{span}(S)$.

Then $\text{im}(T) \overset{(=)}{=} \text{span}(T(S)) \overset{\text{def}}{=} \text{span}(T(v_1), \dots, T(v_n))$.

Pf: ($\supseteq$): $\forall \sum_{i=1}^{n} a_i T(v_i) \in \text{span}(T(S))$,

$\sum a_i T(v_i) \overset{T \text{ lin}}{=} \sum T(a_i v_i) \overset{\substack{T \text{ lin} \\ \text{Exercise}}}{=} T(\sum a_i v_i) \in T(V) = \text{im}(T)$. ▨

($\subseteq$): $\forall w \in \text{im}(T)$, $\exists\, v \in V$ s.t. $T(v) = w$ by def.

$V = \text{span}(S) \Rightarrow v = \sum a_i v_i$ for some $a_1, \dots, a_n \in \mathbb{R}$

$\Rightarrow w = T(v) = T(\sum a_i v_i) = \sum a_i T(v_i) \in \text{span}(T(S))$. □

Ex 2.10: Consider $T: \mathbb{R}^2 \to \mathbb{R}^2$. $T \in \mathcal{L}(\mathbb{R}^2)$.

$$\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x+y \\ x+y \end{bmatrix}$$ (Exercise)

Then Prop 2.9 via std basis e_1, e_2 of $\mathbb{R}^2$ $\leftarrow$ dim 2

$$\Rightarrow \operatorname{im}(T) = \operatorname{span}(T(e_1), T(e_2)) = \operatorname{span}\left(\begin{bmatrix} 1+0 \\ 1+0 \end{bmatrix}, \begin{bmatrix} 0+1 \\ 0+1 \end{bmatrix}\right)$$

$$= \operatorname{span}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) \quad \text{a } \underline{\text{line}}. \quad (\dim 1)$$

Also, $\ker(T) = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 : \begin{bmatrix} x+y \\ x+y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$

$$= \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 : y = -x \right\} = \operatorname{span}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) \quad \underline{\text{line}} \ (\dim 1)$$

$\Downarrow$

Observation: Given $T \in \mathcal{L}(V,W)$ w/ V f.d., is how are $\dim V$, $\dim \ker(T)$, $\dim \operatorname{im}(T)$ related?

(Ex 2.10: guess is $\dim V = \dim \ker T + \dim \operatorname{im} T$

$\Downarrow$ Yes!

Def 2.11: $T \in \mathcal{L}(V,W)$.

(1) nullity of T is $\operatorname{nullity}(T) := \dim \ker(T)$.

(2) rank (rk) of T is $\operatorname{rank}(T) := \dim \operatorname{im}(T)$.

$\Downarrow$

Thm 2.12 (Rank-Nullity Thm) (R-N Thm) (Dim Thm in FIS)

$T \in \mathcal{L}(V,W)$. Then $\boxed{\dim V = \operatorname{nullity}(T) + \operatorname{rank}(T)}$.

Pf of R-N when V f.d.:

V f.d. $\Rightarrow$ $\ker(T)$ f.d. b/c of Prop 2.6 (& §1.4).

Choose a basis $\beta_0 = \{v_1, \dots, v_k\}$ of $\ker(T)$ (so $k = \text{nullity}(T) < \infty$). Then by BET, extend β_0 to a basis $\beta := \{v_1, \dots, v_k, v_{k+1}, \dots, v_n\}$ of V (so $\dim V = n \geq k$). $V = \text{span}(\beta)$ so Prop 2.9

$\Rightarrow \text{im}(T) = \text{span}(T(\beta))$

$= \text{span}(\underbrace{T(v_1), \dots, T(v_k)}_{0}, \underbrace{T(v_{k+1}), \dots, T(v_n)}_{n-k \text{ elts}})$

$\Rightarrow \text{rk}(T) \leq n-k = \dim V - \text{nullity}(T)$.

WTS: $T(v_{k+1}), \dots, T(v_n)$ in $\text{im}(T)$ is l.i.

(WTS $\Rightarrow \text{rk}(T) \geq n-k$ by §1.4)

Pf of WTS in next class!